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Operations Research 415
Part IV: Queuing Theory

Lecture 27: The Queuing System $M/M/1/GD/\infty/\infty$

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Birth & Death Process Transitions (pp. 1073–1074)

$$\begin{aligned}
\lambda &= \text{average \# of arrivals in the system per unit time} \\
\mu &= \text{average \# of departures from the system per unit time} \\
L_q &= \text{average \# of customers waiting in line} \\
L_s &= \text{average \# of customers in service} \\
L &= \text{average \# of customers present in the system}
\end{aligned}$$

Then

$$\begin{aligned}
L &= \frac{\rho}{1-\rho} \\
L_q &= L - \rho = \frac{\rho^2}{1-\rho} \\
L_s &= \rho
\end{aligned}$$

Here $\rho = \frac{\lambda}{\mu}$ and note that $\boxed{L = L_q + L_s}$

Little's Queuing Formulae (pp. 1074–1075)

$$\begin{aligned}
\lambda &= \text{average \# of arrivals in the system per unit time} \\
\mu &= \text{average \# of departures from the system per unit time} \\
L_q &= \text{average \# of customers waiting in line} \\
L_s &= \text{average \# of customers in service} \\
L &= \text{average \# of customers present in the system} \\
W_q &= \text{average time a customers spends waiting in line} \\
W_s &= \text{average time a customers spends in service} \\
W &= \text{average time a customers spends in the system}
\end{aligned}$$

Theorem 3: Little's Queuing Formulae (p. 1075)

For any queuing system in which a steady-state distribution holds, $\boxed{L = \lambda W}$, $\boxed{L_q = \lambda W_q}$ and $\boxed{L_s = \lambda W_s}$.

Example 3 (p. 1076)

An average of 10 cars per hour arrive at a single-server drive-in bank teller.

Assume that the average service time for each customer is 4 minutes, and that both inter-arrival times and service times are exponentially distributed.

1. What is the probability that the teller is idle?
2. What is the average number of cars waiting in line for the teller?
3. What is the average amount of time a drive-in customer spends in the bank parking lot (including time in service)?
4. On the average, how many customers per hour will be served by the teller?